METARET – finding papers and data with risk elicitation tasks

Method, choices, sources and results of the first Claude exploration run

Claude (autonomous run requested by Paolo Crosetto)

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Goal

Identify papers that elicited risk attitudes with at least one risk elicitation task (RET) and that have an online data repository; for each, create a folder in `DATA ENTRY/claude exploration/` with a README (bibliography, notes), the paper PDF when open access, and the repository's data and documentation. Scope given by the user: any RET (Holt–Laury, Eckel–Grossman, Gneezy–Potters, BRET, Tanaka–Camerer–Nguyen, Andreoni–Sprenger convex budgets, others), journals JRU, Experimental Economics, JEBO, JESA, agricultural economics (ERAE, Q Open), plus GitHub, AEA registry and "elsewhere". The user then left, asking me to choose sensibly, log choices, and to start small ("a few at a time is fine, as long as the method is sound and we can rerun it").

Method

The pipeline is deliberately API-based so that it can be rerun. Scripts are in `_scripts/`.

1. **Candidate papers** (`oa.py`). OpenAlex full-text search over 15 RET phrases ("risk elicitation", "Holt and Laury", "Eckel and Grossman", "Gneezy and Potters", "bomb risk", "Tanaka", "multiple price list", "DOSPERT", "investment game", …) restricted to six ISSNs: JRU, Experimental Economics, JEBO, JESA, JBEF and ERAE, articles since 2008. Result: 639 candidate articles (JEBO 243, Exp. Econ. 157, JRU 126, JESA 57, JBEF 37, ERAE 19).

2. **Abstract screen** (`cr.py`). Abstracts from Crossref (only 271 of 639 had one), keyword screen on title+abstract: 212 flagged as likely RET users.
3. **Paper → dataset linking, two ways.**
   - DataCite relation search (`dc.py`): datasets listing the paper DOI as related identifier (44 hits, mostly publisher supplements, low yield).
   - DataCite title match (`dct.py`): dataset titles ("Replication data for: …", "Data for: …") fuzzy-matched to the paper title (ratio > 0.8).
4. **Dataset → paper discovery** (`dcs.py`, plus ad hoc Dataverse/Zenodo searches). DataCite full-text search for RET terms in dataset descriptions; I then hand-picked plausible experimental datasets from titles.
5. **Build** (`build.py`, `lib.py`). For each dataset DOI: DataCite metadata → paper DOI (related identifier, else Crossref title match with similarity > 0.85) → Crossref bibliography → Unpaywall open-access PDF (only files that really start with `%PDF`) → dataset download via repository APIs (Harvard/other Dataverse, Zenodo, Mendeley Data, OSF, figshare) with a 250 MB cap → `README.md` and an entry in `_index.csv`.
6. **Vet** (`vet.py`). Counts RET keywords in the paper PDF and in readable data documentation, appended to each README. Indicative only.
7. **Summary** (`summ.py`) regenerates the top-level `README.md` table.

Methodological choices (and why)

- **Any single RET qualifies**, as instructed; within-subject pairs are *not* required at this stage, but are what METARET ultimately needs, so each dataset must still be checked for them.
- **Public data only.** I did not create accounts, accept terms, use tokens or bypass bot challenges. Consequently openICPSR/ICPSR (login), Dryad (token), UK Data Service (registration) and Radboud RDR datasets are recorded with metadata and link but **not downloaded**: they are folders marked "no (manual)". heiDATA first served a "not a bot" challenge to a browser-like User-Agent; I did not circumvent it and instead used an honest script User-Agent, which was accepted.
- **250 MB cap** per dataset to keep the Dropbox folder manageable; one dataset (Pahontu 2020, 491 MB) was skipped for this reason.
- **Data left as published**, not cleaned or recoded; cleaning follows the existing `clean_data.R` convention in `Data/` later.
- **Conservative automatic matching.** Title similarity thresholds (0.8 for discovery, 0.85 for bibliographic matching). One wrong match (a Zenodo urn-task dataset matched to an unrelated IEEE paper) was caught on inspection and the folder was relabelled "paper not identified". A keyword false positive (Brañas-Garza et al. 2023: time preferences only, no RET) was moved to `_excluded/`.
- **Overlap with METARET not yet checked.** The README of the repository lists ~90 papers already included; I did not systematically remove overlaps, but none of the folders below matches a title on that list as far as I could see (the README list was read once; a formal check by DOI/bibkey against `METARET.bib` is a to-do).
- **Honest labelling.** Where a paper could not be confirmed to contain a RET automatically, the table says "(not verifiable automatically)" rather than asserting it.

Sources used

OpenAlex (candidate discovery, until its free daily budget ran out), Crossref (bibliography, abstracts), DataCite (dataset discovery and metadata), Unpaywall (open-access PDFs), Harvard Dataverse, DataverseNL, heiDATA, Zenodo, Mendeley Data, OSF, figshare (data), and a few web searches (little value; the search tool returned mostly paper landing pages). GitHub repository search turned up only oTree/Redwood implementations of Holt–Laury and BRET (e.g. `felixholzmeister/bret`, `Leeps-Lab/redwood-holt-laury`), not paper datasets; they are not downloaded but are useful for the "Tasks" pages of the app.

Results

36 candidate folders were created (plus one excluded). Of these, 23 have data downloaded, 13 are metadata-only (login, token, size cap or no files listed), and 6 have a paper PDF. Total size 170 MB. The notable leads for METARET are:

- **Tanaka, Camerer & Nguyen (2010, AER)** – the "Vietnam data Camerer" lead already in `DATA ENTRY`; openICPSR 112336, login required.
- **Andreoni & Sprenger (2012, AER)** and the two AER comments (Cheung 2015; Miao & Zhong 2015) – risk vs. time (convex budgets and uncertainty equivalents); openICPSR, login required.
- **Sutter et al. (2013, AER)** – risk/time experiments with adolescents linked to field behaviour (a natural RET-vs-real-behaviour case); openICPSR.
- **Downloaded now:** Ergin et al. 2025 (Fork Game, higher-order risk), Gächter et al. 2021 (loss aversion in riskless and risky choices), Huber et al. 2023 (risk aversion in wealth, field), Müller & Rau 2016 (risk and other-regarding preferences), Crockett & Crockett 2019 (BDM and Holt–Laury), Halko et al. 2021, Gross et al. 2021, Haile et al. 2020, Brunner et al. 2014, Mudzingiri 2024, Woike et al. 2026, Gallenstein 2024, Kassas et al. 2021 and several datasets whose paper could not be identified automatically.

Folder	Data	PDF	RET keywords (automatic)
AlsFerrer_2023_Experimental	yes	yes	Tanaka/CPT, Lotteries/gambles, Self-report, Convex Time Budget
Andreoni_2012_AmericanEcon	no (manual)	no	(not verifiable automatically)
Blavatskyy_2020_SSRNElectron	no (manual)	no	(not verifiable automatically)
Brunner_2014_GamesandEcon	yes	no	(not verifiable automatically)
Cheung_2015_AmericanEcon	no (manual)	no	(not verifiable automatically)
Crockett_2019_JournalofEco	yes	no	(not verifiable automatically)
Ergin_2025_Experimental	yes	yes	Holt-Laury, Certainty equiv./BDM, Lotteries/gambles, Self-report
Fllbrunn_2018_EconomicInqu	no (manual)	no	(not verifiable automatically)
Gallenstein_2024_JournalofRis	yes	no	(not verifiable automatically)
Gchter_2021_TheoryandDec	yes	yes	Holt-Laury, Certainty equiv./BDM, Lotteries/gambles
Gross_2021_Psychologica	yes	no	(not verifiable automatically)
Haile_2020_WorldDevelop	yes	no	(not verifiable automatically)
Halko_2021_JournalofEco	yes	no	(not verifiable automatically)
Hoel_2022_Journalofthe	no (manual)	no	(not verifiable automatically)
Huber_2023_Experimental	yes	yes	Holt-Laury, Gneezy-Potters, Lotteries/gambles, Self-report
Incentivized_binary_choice_under_risk_____zenodo.777	yes	no	Lotteries/gambles

Folder	Data	PDF	RET keywords (automatic)
Kassas_2021_PLOSONE	yes	yes	Gneezy-Potters, Lotteries/gambles, Self-report
Miao_2015_AmericanEcon	no (manual)	no	(not verifiable automatically)
Mller_2016_EuropeanEcon	yes	no	(not verifiable automatically)
Mudzingiri_2024_SageOpen	yes	no	(not verifiable automatically)
Murad_2016_JournalofRis	no (manual)	yes	Holt-Laury, Gneezy-Potters, Certainty equiv./BDM, Lotteries/gambles, Self-report
Noussair_2013_TheReviewofEno	(manual)	no	(not verifiable automatically)
Pahontu_2020_SSRNElectron	no (manual)	no	(not verifiable automatically)
Raw_Data_Loss_Aversion_Wyk	2K N1M	no	(not verifiable automatically)
Reallife_financial_risk_taking	yes	study_I1BR2J	no	(not verifiable automatically)
Risk_Attitudes_and_Microfinance	no (manual)	Profi	en_zjk4ysbmg6	no	(not verifiable automatically)
Risky_Choice_in_Virtual_Reality	yes	_Predicts_m9.figshar	no	(not verifiable automatically)
SelfInfliction_of_Pain_Increases	yes	_Unreal_BITSMP	no	BRET, Lotteries/gambles, Self-report, Holt-Laury
Sutter_2013_AmericanEcon	no (manual)	no	(not verifiable automatically)
Tanaka_2010_AmericanEcon	no (manual)	no	(not verifiable automatically)
Urn_risk_ambiguity_dataset_ye	s	nodo22116575	no	Certainty equiv./BDM, Lotteries/gambles
Vainio_Heinonen_2022_x2k2ky8	5kg	no	(not verifiable automatically)
Woike_2026_JournalofBeh	yes	no	Lotteries/gambles, Self-report
of_impulsivity_and_risk_prefe	rences_A_q_f4fknw2vfn	no	(not verifiable automatically)
revealed_vs_perceive_risk_pre	ferences_hrcdhf6y82	no	(not verifiable automatically)
vanRooij_2007_JournalofPub	no (manual)	no	(not verifiable automatically)

Limitations

- **Recall is low.** The 212 flagged papers in the six journals mostly have *no* dataset findable through DataCite; many journals host data as publisher supplements or on author sites not indexed there. Springer/Elsevier/Cambridge landing pages and the AEA registry (not an API I could use) were not scraped. The result is a lead list, not a census.
- **Precision is unverified for several folders.** The RET keyword check works only when the PDF is present or data documentation is textual; binary data (xlsx, dta, sav) are not inspected.
- **OpenAlex outage** (free budget exhausted by my early requests) prevented abstract-based screening through OpenAlex and the planned extension to ERAE, Q Open and other journals beyond the first pass (ERAE yielded only 19 candidates; Q Open and JEP returned none).
- **Early use of the user's email address** in OpenAlex `mailto` parameters: this should not have been done; no personal email was used afterwards (Unpaywall uses a public test address).

How to rerun or extend

1. `python3 oa.py` (needs an OpenAlex API key or a fresh daily budget) → `cr.py` → `dct.py` / `dcs.py`: new candidates.
2. Pick DOIs and run `python3 build.py <dataset-doi> ...`: builds folders; it appends to `_index.csv`.
3. `python3 vet.py` and `python3 summ.py` refresh keyword checks and the top-level README.
4. Re-render this report: `quarto render report.qmd`.

Suggested next steps

1. Manually download the openICPSR/Dryad packages (Tanaka et al., Andreoni–Sprenger, Sutter et al., Hoel et al., Blavatskyy et al., Füllbrunn et al., Noussair et al., van Rooij et al.).
2. Read the READMEs of the "paper not identified" folders and fix their bibliography.
3. Check each downloaded dataset for within-subject RET pairs and ask: is it new relative to the METARET list?
4. Extend discovery to ERAE/Q Open/Agricultural Economics and to AEA registry/OSF registrations; consider mining reference lists of existing METARET papers and the `METARET.bib` for datasets.